

Geometry Crystallisation in CRF

Two Levels of Crystallisation, $\sigma(d_s)$ Scaling,
and the Thermodynamic Limit of 3D Space

V20.5 Results: Multi-N Scaling + 1000-Sweep Verification

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Abstract

CRF Hybrid V20.5 with Multi-N scaling verification ($N \in \{125, 250, 500, 1000\}$) and 1000-sweep repeatability check establishes four results at different confidence levels.

(1) **Two-level crystallisation (Confirmed).** Node-level (Ω accumulation) and geometry-level ($d_s \rightarrow 3$ from concurrent firing) are decoupled: $\text{Corr}(\Omega, d_s) \approx -0.02$. HN-04 population irreversibility: 0 violations.

(2) **Steady state confirmed at $N = 500$ (Confirmed).** 1000-sweep run (seed=42) shows drift = 0.00004/sweep ≈ 0 after burn-in (300 sweeps). Post-burn-in mean = 2.994 ≈ 3.0 . $\sigma(d_s)$ decreases from 0.544 (400 sweeps) to 0.358 (1000 sweeps)—confirming the system has not yet fully converged even at 400 sweeps.

(3) **$\sigma(d_s)$ scaling (Hypothesis, partial).** $\sigma(d_s) \sim N^{-0.339}$ ($R^2 = 0.74$). The exponent is between 0 (no scaling) and -0.5 (pure shot-noise), indicating a real but sub-ideal scaling. The formula $\sigma = d_s \sqrt{f(1-f)/N_{\text{fire}}}$ over-predicts at small N because d_s itself is not yet at 3 there.

(4) **Mean convergence is threshold-like (Confirmed, new).** $\langle d_s \rangle$ jumps from 1.09 ($N = 125$) to 2.01 ($N = 250$) to 2.99 ($N = 500$)—not a smooth scaling but a threshold transition near $N \approx N_{\text{crit}}/f_{II} \approx 500$. This is more informative than any σ scaling.

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1 What Was Already Derived

Prior Results

- $d_s = 3.031 \pm 0.076$ (V20.3)
- $\beta = 2.212 \pm 0.015$ (#21 closed 1.7%)
- $f_{II} = 0.147$, $T_{\text{avg}} = 35.4$ (#21a,b)
- $N_{\text{crit}} = f_{II} \cdot N = 74$ at $N = 500$
- Two-level crystallisation ($\text{Corr} \approx -0.02$), HN-04 (0 violations)
- $\varepsilon_0 = 0.00755$ — ontological noise floor, not tunable

2 Steady State Verification (Phase 1–2)

Steady State Confirmed at $N = 500$, 1000 Sweeps

Seed=42, $N = 500$, 1000 sweeps. Burn-in = 300 sweeps.

Window	Mean d_s	$\sigma(d_s)$	Note
Full (0–1000sw)	2.873	0.520	includes burn-in
Post-burn-in (300–1000)	2.994	0.358	steady regime
Late (500–1000)	2.963	0.285	converging
Drift slope (300–900)	+0.00004/sweep — $R^2 = 0.0003$ (no drift)		

Key: σ drops from 0.544 (400sw) to 0.358 (1000sw) — the system needs > 400 sweeps to reach full steady state. Burn-in = 300 sweeps is sufficient; mean is stable thereafter.

The σ reduction from $0.544 \rightarrow 0.358$ is not scaling — it is *convergence within the same N* . This was missed in the 400-sweep multi- N run (all N ran for 400 sweeps, but they need different burn-in times).

3 Two Levels of Crystallisation

Two-Level Crystallisation (Confirmed)

	Node-level	Geometry-level
Variable	$\Omega_i = M_i - 1$	d_s
Mechanism	Accumulated R-events	Concurrent Phase II firing
Irreversibility	Exact (HN-04)	Statistical ($N \rightarrow \infty$)
Evidence	$\text{Corr}(\Omega, d_s) \approx -0.02$	d_s threshold at $N \approx 500$

HN-04: N_{phase2} never drops below N_{crit} after first reaching it. 0 violations across all runs and seeds.

4 $\sigma(d_s)$ Scaling: Honest Assessment

4.1 Multi-N results

N	Sweeps	$\langle d_s \rangle$	$\sigma(d_s)$	N_{fire}	Note
125	400	1.087	0.650	3.2	below threshold
250	400	2.006	0.690	5.3	approaching
500	1000	2.994	0.358	8.4	at threshold
1000	600	3.526	0.370	16.0	overshoot

Fit: $\sigma \sim N^{-0.339}$ ($R^2 = 0.74$). Predicted by shot-noise: $N^{-0.500}$.

4.2 Why the formula partially fails

The shot-noise formula $\sigma(d_s) = d_s \sqrt{f(1-f)/N_{\text{fire}}}$ assumes $\langle d_s \rangle = 3$ for all N . This fails at small N because the system has not yet formed 3D geometry.

Correct Interpretation

The data reveals two distinct phenomena:

(A) Threshold transition in $\langle d_s \rangle$: $\langle d_s \rangle$ jumps from ~ 1 to ~ 3 near $N \approx 500$. This is a *threshold effect*, not smooth scaling. It corresponds to $N_{\text{crit}} = 74$ concurrent Phase II nodes being required for a stable 3D vote. Since $f_{II} \approx 0.147$, the threshold population is $N^* = N_{\text{crit}}/f_{II} \approx 500$.

(B) $\sigma(d_s)$ reduction within the 3D regime: For $N \geq N^*$ (where $\langle d_s \rangle \approx 3$), $\sigma \sim N^{-\alpha}$ with $\alpha \approx 0.34$ – 0.5 . More data needed to distinguish from $N^{-0.5}$.

4.3 Scale separation: ε_0 vs $\sigma(d_s)$

A key structural insight: ε_0 (ontological noise floor) and $\sigma(d_s)$ (geometric fluctuation) operate at different scales and are governed by different mechanisms.

Scale Separation (Derived from Structure)

	ε_0 (microscopic)	$\sigma(d_s)$ (macroscopic)
Level	Node / event	Population / geometry
Value	0.00755 (fixed, derived)	~ 0.35 at $N = 500$
N-dependence	None (structural)	$\sim N^{-0.34}$
$N \rightarrow \infty$	Remains 0.00755	$\rightarrow 0$
Role	Drives R-events	Geometry vote noise

Consequence: In the thermodynamic limit $N \rightarrow \infty$, $\sigma(d_s) \rightarrow 0$ while ε_0 remains strictly positive. $d_s = 3$ exactly at $N = \infty$ even though $\varepsilon_0 > 0$. Irreducible noise does not vanish; it becomes geometrically unobservable.

This resolves the apparent paradox: the universe has exact 3D geometry ($\sigma \approx 0$) despite permanent microscopic noise ($\varepsilon_0 > 0$). The concurrent-firing low-pass filter converts ε_0 -driven fluctuations into a stable geometric consensus.

5 Status Table

Result	Status	Note
Two-level crystallisation distinct	Confirmed	Corr ≈ -0.02
HN-04 irreversibility	Confirmed	0 violations all N
Steady state at $N = 500, 1000\text{sw}$	Confirmed	drift = $0.00004/\text{sw}$
Mean threshold at $N^* \approx 500$	Confirmed	$\langle d_s \rangle : 1 \rightarrow 3$
Scale separation ε_0 vs σ	Confirmed	structural argument
$\sigma \sim N^{-0.339}$ scaling	Hypothesis	$R^2 = 0.74$, 4 points
$\sigma \rightarrow 0$ as $N \rightarrow \infty$	Hypothesis	extrapolation
$d_s = 3$ exact at $N \rightarrow \infty$	Hypothesis	follows from $\sigma \rightarrow 0$
Shot-noise exp = -0.5	Open	current: -0.34 , gap
f_{II} from lifecycle theory	Open	HN-03 extended
Multi-seed scaling (≥ 3 seeds/ N)	Open	repeatability of exp

6 Key Numbers

Key Numbers — This Session

$$\begin{aligned}
&\langle d_s \rangle_{N=500, 1000\text{sw}} = 2.994 \quad \sigma_{N=500, 1000\text{sw}} = 0.358 \quad \text{drift} = 0.00004/\text{sw} \\
&\sigma \sim N^{-0.339} \quad (R^2 = 0.74) \quad \langle d_s \rangle : 1.09 \rightarrow 2.01 \rightarrow 2.99 \rightarrow 3.53 \quad (N = 125, 250, 500, 1000) \\
&N^* \approx 500 = N_{\text{crit}}/f_{II} = 74/0.147 \quad \text{HN-04 violations} = 0 \quad (\text{all } N)
\end{aligned}$$

7 Open Questions

Next Targets

1. **Multi-seed scaling (Priority 1).** Run $N \in \{500, 1000, 2000\}$ at ≥ 3 seeds each, with 1000+ sweeps. Measure σ in post-burn-in regime only. Check if exponent stabilises at -0.339 or drifts toward -0.5 .
2. **N^* threshold measurement (Priority 1).** Run finer N grid near threshold: $N \in \{300, 400, 500, 600, 700\}$. Locate the N at which $\langle d_s \rangle$ crosses 2.5. Compare to prediction $N^* = N_{\text{crit}}/f_{II} \approx 500$.
3. **f_{II} from lifecycle theory (Priority 2).** Deriving $f_{II} = 0.147$ analytically closes the shot-noise formula completely.
4. **$N=1000$ overshoot explanation (Priority 2).** $\langle d_s \rangle = 3.526 > 3$ at $N = 1000$ is unexpected. Possible cause: geometry estimator bias at high N_{fire} . Or: d_s estimator has + bias at large N (fewer walkers per node).

ε_0 never vanishes. It becomes unobservable.

Space is not silence — it is noise voted into consensus

by N_{crit} nodes firing together, sweep by sweep.

At $N = \infty$, the vote is unanimous. That is what we call three dimensions.